

Example: /functional-unit wool_yarn (v1 — with structure graph and reference process)

This is a recorded example of the functional-unit teaching skill in action, using the wool_yarn case study from skills_references/wool_yarn/.

This version demonstrates the updated skill flow where the structure diagram and reference process are introduced automatically in Step 2 — without the student having to ask for them.

The conversation

AI: A knitwear brand wants to put a carbon footprint label on their wool yarn so buyers can compare it to other fibres.

What would you measure the footprint per — per kilogram of yarn, per sweater, or per year of wearing?

Student: Per kilogram of yarn.

AI: That's a great instinct — and it's actually exactly what this study uses. Per kilogram is the natural choice when you're measuring a raw material at the point it leaves the mill, because it lets any buyer compare two yarn suppliers on equal footing, regardless of what they plan to make with it.

The study we'll be working through is called **Wool Yarn — 1 kg**. It traces the climate impact of producing 1 kg of wool yarn, following the supply chain from the sheep farm through scouring and spinning at the mill. It's a teaching example built for this course — the numbers are illustrative rather than from a specific brand, but calibrated to be realistic.

Here's the supply chain diagram:

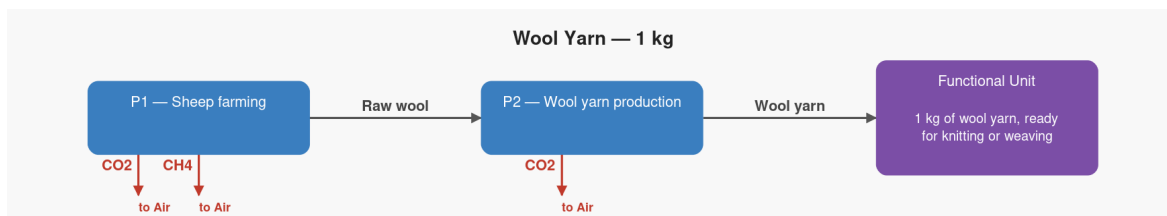


Figure 1: Wool yarn supply chain — structure

The rightmost box — **P2, Wool yarn production** — has a special name in LCA: it's called the **reference process**. That just means it's the finishing line — the step that delivers the finished product. Everything to its left (the sheep farm) exists only to supply it. The functional unit is precisely what comes out of that box: the thing we're measuring, and the point at which the study stops.

So in LCA, this precise definition — **1 kg of wool yarn, ready for knitting or weaving** — is called the **functional unit**. It rules out measuring per sweater, for example, because different sweater designs use different amounts of yarn, which would make comparisons unfair.

Here's a what-if to push the idea further: if a competitor yarn supplier measured their footprint **per sweater** instead of per kilogram, and their yarn happened to be thicker so each sweater used more of it — could they make their yarn look better or worse than it really is?

Student: Yes, but I am just guessing sort of.

AI: That's a perfectly honest answer — and the instinct is right. Let's make it concrete so it clicks properly.

Say Supplier A's yarn has a footprint of **15 kg CO₂ per kg of yarn**. Supplier B's yarn has a footprint of **18 kg CO₂ per kg of yarn** — genuinely worse. But Supplier B's yarn is thicker, so a sweater only needs 400 g of it, while Supplier A's yarn needs 600 g per sweater.

Now watch what happens if both report **per sweater**: - Supplier A: $15 \times 0.6 =$ **9 kg CO₂ per sweater** - Supplier B: $18 \times 0.4 =$ **7.2 kg CO₂ per sweater**

Supplier B now looks better — even though their yarn is dirtier per kilogram. The unit switch has completely reversed the comparison, just by hiding the fact that the sweaters use different amounts of yarn.

This is exactly why choosing the right functional unit matters so much. When a supplier sends you a sustainability number, the first thing to check is always: **per what?** A number without a unit isn't a claim — it's just a number.

As a buyer or yarn sourcing manager, knowing this lets you ask the right follow-up question the moment a supplier hands you a data sheet: "Is that per kilogram of yarn, or per finished garment?" That one question tells you whether the comparison is fair.

Would you like to explore this with more examples, or shall we move on to look at the wool yarn supply chain in more detail?

Student: Yes, can I see more examples?

AI: Here are three that come up often in fashion and retail, each showing a different version of the same trap.

Fast fashion vs quality

Brand A sells a £15 T-shirt, footprint 2 kg CO₂, but it pills after 10 washes and gets thrown away. Brand B sells a £60 T-shirt, footprint 4 kg CO₂, that lasts 200 washes over three years.

- Measured **per garment**: Brand B looks twice as bad.
- Measured **per wear**: Brand A = $2 \div 10 = \mathbf{0.20 \text{ kg CO}_2}$. Brand B = $4 \div 200 = \mathbf{0.02 \text{ kg CO}_2}$. Brand B is ten times better.

The wrong unit here is per garment — it hides the durability difference completely, which is the whole point of the comparison.

Cotton tote bag vs disposable paper bag

Measured per bag produced, a cotton tote has a footprint of ~7 kg CO₂ versus ~0.04 kg CO₂ for a paper bag — the tote looks 175 times worse. But used 200 times, the tote's per-use footprint drops to 0.035 kg CO₂ — lower than a single paper bag.

The wrong unit here is per bag produced for a product designed to be reused. A cradle-to-gate number always makes reusable products look bad compared to disposables.

Chiffon vs denim fabric

A fabric supplier claims their chiffon (80 g per square metre) has a lower carbon footprint than denim (400 g per square metre), measured per square metre. They're right — but a dress made from chiffon uses far more metres of fabric than a pair of jeans. So the per-square-metre comparison is misleading when the two products require very different amounts of material.

Per kilogram would be fairer for comparing raw materials; per finished garment would be fairer if you want to compare actual products.

The pattern is always the same: the unit that makes a product look good tends to be the one that hides the most important fact.

Student: Yes, let's go on.

AI: That wraps up the functional unit lesson — you’ve worked through all the key ideas. To summarise in one sentence: **a sustainability number means nothing without knowing precisely what it’s measuring, and the functional unit is that definition.**

The natural next step is the supply chain lesson, which goes deeper into that diagram you saw — walking through what each box represents in the real world, what flows along the arrows, and what the study boundary includes and leaves out. You’d invoke it with:

/supply-chain wool_yarn

Want to jump straight into that, or is there anything about functional units you’d like to come back to first?

What this example demonstrates

- The student answered the opening question correctly (per kilogram) — the skill validated this warmly and confirmed it was the right choice before moving on, rather than treating a correct answer as a setup for a correction
- The case study introduction, structure diagram, and reference process all appeared in the same response, giving the student three things at once: context (what the study is), a visual (the supply chain), and vocabulary (reference process) — before the functional unit was named
- “Reference process” was introduced and defined in plain English immediately: “the finishing line — the step that delivers the finished product”
- The what-if question used yarn-specific numbers (Supplier A 600 g per sweater, Supplier B 400 g per sweater) to show how the per-sweater unit reverses the comparison — the maths was shown step by step so the student could follow it
- The student said “I am just guessing sort of” — an honest hedge — which was validated before the concrete example was given, so they didn’t feel wrong
- Three wrong-unit examples (fast fashion vs quality, tote vs paper bag, chiffon vs denim) were provided when asked, using the reference table in the skill
- The lesson closed by naming the next skill (/supply-chain wool_yarn) so the student has a clear path forward